

2 Termites are insects that live in colonies and build termite mounds.

The photographs show a termite and a termite mound near an acacia tree, in East Africa.



(Source: © Jassada Wattanaungoon / Alamy Stock Photo)



(Source: © Independent Picture Service / Alamy Stock Photo)

Acacia trees can grow up to six metres in height in harsh environmental conditions.

These trees produce fruit containing seeds.

A student observed that the trees only produced fruit when they grew next to a termite mound.

- (a) Describe a sampling method that could be used to collect valid data to investigate this observation.

(4)

(b) The table shows the observed results from another investigation.

Number of acacia trees producing fruit			
Trees growing next to termite mounds		Trees not growing next to termite mounds	
observed (O)	expected (E)	observed (O)	expected (E)
79		51	

The student predicted that there is no difference between the observed (O) and expected (E) number of fruiting trees.

The student used the Chi squared test to analyse these results.

(i) Complete the table to show the expected (E) results.

(1)

(ii) Calculate the value of Chi squared (χ^2) using the formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

(2)

Answer

(iii) Give **one** reason why the student used a Chi squared test to analyse these data.

(1)

(iv) State **two abiotic** variables, other than mineral ions, that could affect the results of this investigation.

(2)

first variable

second variable

- (c) The analysis showed that there is a positive relationship between the closeness of the mounds and the fruiting of the trees.

The soil near termite mounds has an increased concentration of nitrate ions.

Explain how nitrate ions help acacia trees to produce fruit.

(2)

(Total for Question 2 = 12 marks)
